LESSON 4.2

Relate Fractions, Decimals, and Percentages

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write equivalent fractions, decimals, and percents for the same value.

WHY THIS MATTERS You're designing the scoring system for a new arcade game. Players earn points as fractions of the total possible score. The leaderboard shows scores as percentages, but the game engine uses decimals. You need to convert between all three forms to build the display! A player scored 3/4 of the total possible points. Write that score as a decimal and as a percent.

REMEMBER

Relate Fractions, Decimals, and Percentages

Fraction, Decimal & Percent Equivalence Equivalent forms name the same value in different ways. Percent means "out of 100," so a fraction is already a percent once its denominator is 100.

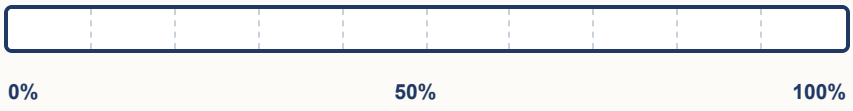
1. Percent to Decimal: Divide by 100 (move decimal point 2 places left).
2. Decimal to Percent: Multiply by 100 (move decimal point 2 places right).
3. Fraction to Percent: Divide numerator by denominator, then convert decimal to percent

Words: **Percent *(Porcentaje)*** **Decimal *(Decimal)*** **Equivalent *(Equivalente)*** **Benchmark *(Referencia)*** **Discount *(Descuento)***

1 **Circle the letter of the best answer. Show how you know.**

On a 100-square grid, 37 squares are shaded. What percent of the grid is shaded?

(A) 0.37%
(B) 3.7%
(C) 37%
(D) 63%



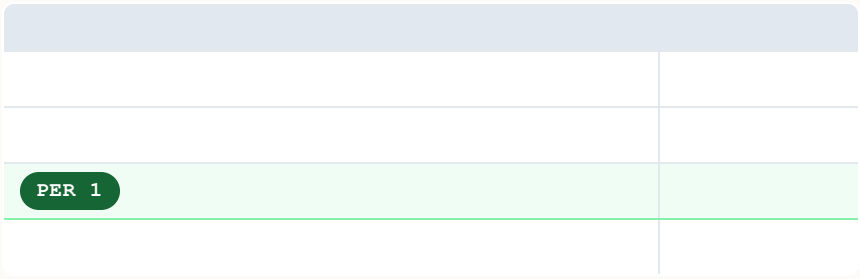
WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

2 **Circle the letter of the best answer. Show how you know.**

Percent means 'per ___'.

(A) 1
(B) 10
(C) 100
(D) 1,000



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

A tape diagram is split into 4 equal parts and 3 parts are shaded. What percent is shaded?

(A) 30%
(B) 34%
(C) 43%
(D) 75%

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A price rises to 150% of what it was. Compared with the original price, the new price is…

(A) half as large
(B) exactly the same
(C) one and a half times as large
(D) 150 times as large

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A store has a clearance sale: 'Everything 3/5 off!' Your friend says that means 35% off. Another friend says it's 60% off. The original price of a game is $40. '3/5 off' means 3 of the 5 equal parts come off the price. Which friend is right?

(A) The friend who said 60% off
(B) The friend who said 35% off
(C) Both are the same amount
(D) Neither — 3/5 is not a percent

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A store has a clearance sale: 'Everything 3/5 off!' Your friend says that means 35% off. Another friend says it's 60% off. The original price of a game is $40. Write 3/5 as a decimal.

(A) 0.35
(B) 0.53
(C) 0.6
(D) 3.5



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

EXPLAIN YOUR THINKING

What number did you multiply by to turn each denominator into 100, and how did that fraction out of 100 give you the percent?

SENTENCE STARTER I multiplied the top and bottom of ___ by ___ to get ___/100, so the percent is ___%.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each arcade score tile to its equal fraction, decimal, and percent.

_____ 4/5	(A) 0.8 = 80%
_____ 9/10	(B) 0.9 = 90%
_____ 1/2	(C) 0.5 = 50%
_____ 1/4	(D) 0.25 = 25%

2 **Write the name of the correct group on each line.**

Sort each arcade tile into the form it is already written in.

Groups: Fraction tiles | Decimal tiles | Percent tiles

_____ 2/5

_____ 7/10

_____ 0.3

_____ 0.85

_____ 70%

_____ 15%

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A store has a clearance sale: 'Everything 3/5 off!' Your friend says that means 35% off. Another friend says it's 60% off. The original price of a game is $40. The game costs $40 and 3/5 comes off. How much is taken off?

(A) $14
(B) $16
(C) $24
(D) $35



WORK

4 **Circle the letter of the best answer. Show how you know.**

A store has a clearance sale: 'Everything 3/5 off!' Your friend says that means 35% off. Another friend says it's 60% off. The original price of a game is $40. So what do you actually pay for the game?

(A) $16
(B) $24
(C) $26
(D) $40



WORK

5 **Circle the letter of the best answer. Show how you know.**

Your arcade scoreboard reads 0.25 of the level cleared. What percent and fraction match this score?

(A) 2.5% and 1/4
(B) 25% and 1/4
(C) 25% and 1/5
(D) 250% and 1/40

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A player powered up 3/5 of the energy bar. Which decimal and percent name the same amount?

(A) 0.35 and 35%

(B) 0.6 and 60%

(C) 0.6 and 6%

(D) 0.65 and 65%

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What number did you multiply by to turn each denominator into 100, and how did that fraction out of 100 give you the percent?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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